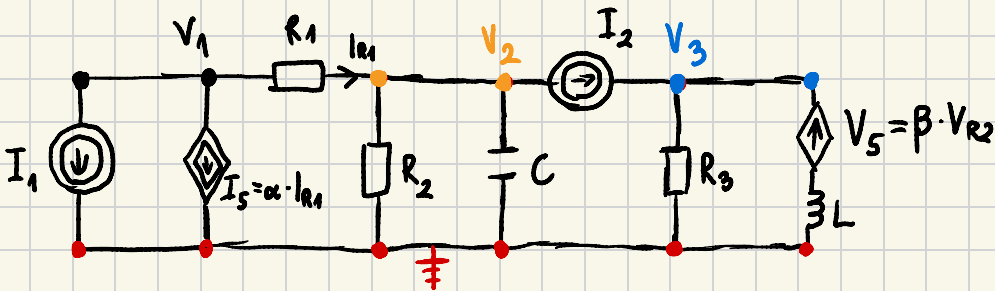


ĆWICZENIA 6

Zbiór zadań - Z. Filipowicz
 ↗ Zygmont

Zad. 1



$$[Y] \cdot [V] = [I]$$

$$I_5 = \alpha \cdot I_{R1} = \alpha Y_1 (V_1 - V_2)$$

$$V_5 = \beta \cdot V_{R2} = \beta V_2$$

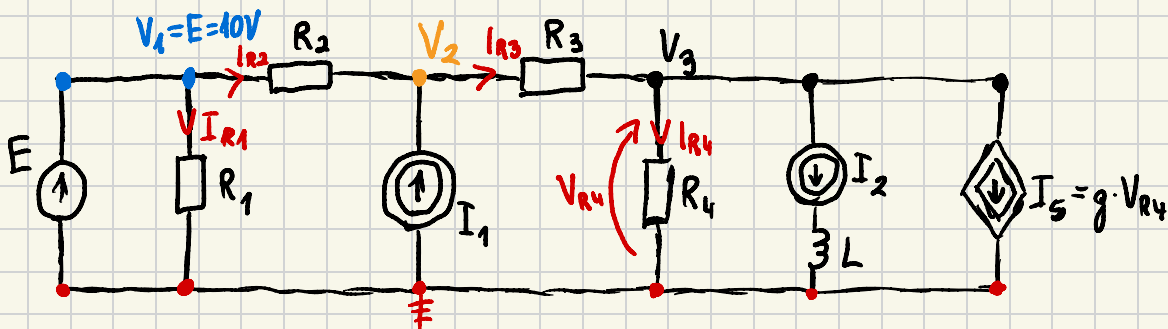
$$\begin{cases} V_1 Y_1 - V_2 \cdot Y_1 = -I_1 - I_5 \\ V_2 (Y_1 + Y_2 + Y_C) - V_1 Y_1 - V_3 \left(\frac{1}{\infty}\right) = -I_2 \\ V_3 (Y_3 + Y_L) - V_2 \left(\frac{1}{\infty}\right) = I_2 + V_5 \cdot Y_L \end{cases}$$

$$\begin{cases} V_1 Y_1 - V_2 Y_1 = -I_1 - \alpha Y_1 V_1 + \alpha Y_1 V_2 \\ V_2 (Y_1 + Y_2 + Y_C) - V_1 Y_1 - V_3 \left(\frac{1}{\infty}\right) = -I_2 \\ V_3 (Y_3 + Y_L) - V_2 \left(\frac{1}{\infty}\right) = I_2 + \beta V_2 Y_L \end{cases}$$

$$\begin{cases} V_1 [Y_1 (1 + \alpha)] - V_2 [Y_1 (1 + \alpha)] = -I_1 \\ -V_1 Y_1 + V_2 (Y_1 + Y_2 + Y_C) - V_3 \left(\frac{1}{\infty}\right) = -I_2 \\ -V_2 \left(\frac{1}{\infty} + \beta Y_L\right) + V_3 (Y_3 + Y_L) = I_2 \end{cases}$$

$$\begin{bmatrix} Y_1(1+\alpha) & -Y_1(1+\alpha) & 0 \\ -Y_1 & Y_1+Y_2+Y_C & 0 \\ 0 & -\beta Y_L & Y_3+Y_L \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} -I_1 \\ -I_2 \\ I_2 \end{bmatrix}$$

Zad. 2



$$e(t) = 10\sqrt{2} \sin \omega t$$

$$E = 10V$$

$$i_1(t) = 4\sqrt{2} \sin(\omega t + 90^\circ)$$

$$I_1 = 4e^{j90^\circ}$$

$$i_2(t) = 10\sqrt{2} \sin(\omega t - 90^\circ)$$

$$I_2 = 10e^{-j90^\circ}$$

$$R_1 = R_2 = R_3 = R_4 = 1 \Omega$$

$$g = 2$$

$$V_2(Y_2 + Y_3) - V_3Y_3 - V_1Y_2 = I_1$$

$$V_3(Y_4 + Y_3 + \cancel{Y_L}) - V_2 \cdot Y_3 = -I_2 - I_5$$

bo patryczne
rozregane z zródiem
prądowym, którego opór
jest nieskończony

$$\begin{cases} V_2(2) - V_3(1) - 10 = j4 \\ V_3(2) - V_2(1) = j10 - 2V_3 \end{cases}$$

$$\begin{cases} 2V_2 - V_3 = 10 + j4 \\ 4V_3 - V_2 = 10j \quad || \cdot 2 \end{cases}$$

$$\begin{cases} 2V_2 - V_3 = 10 + j4 \\ -2V_2 + 8V_3 = 20j \end{cases}$$

$$7V_3 = 10 + 24j$$

$$V_3 = \frac{10}{7} + \frac{24}{7}j = 1,428 + 3,428j$$

$$V_2 = 4\left(\frac{10}{7} + \frac{24}{7}j\right) - 10j = \frac{40}{7} + \frac{96}{7}j - 10j = \frac{40}{7} + \frac{26}{7}j$$

$$V_2 = 5,714 + 3,714j$$

$$V_{2M} = \sqrt{5,714^2 + 3,714^2} = 6,81[V]$$

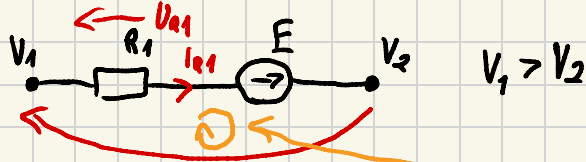
$$I_{R1} = \frac{V_1}{R_1} = \frac{10}{1} = 10[A]$$

$$V_{3M} = \sqrt{1,428^2 + 3,428^2} = 3,71[V]$$

$$I_{R2} = \frac{V_1 - V_2}{R_2} = \frac{10 - 5,714 - 3,714j}{1} = 4,286 - 3,714j[A]$$

$$I_{R3} = \frac{V_2 - V_3}{R_3} = \frac{5,714 + 3,714j - 1,428 - 3,428j}{1} = 4,286 + 0,286j[A]$$

$$I_{R4} = \frac{V_3 - 0}{R_4} = 1,428 + 3,428j$$

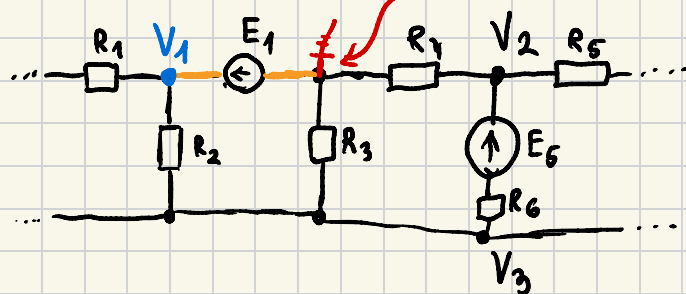


"Wymyślone" oczko, po to
by zastosować napięciowe
prawo Kirchhoffa

$$(V_1 - V_2) - R_1 \cdot I_{R1} + E_1 = 0$$

$$I_{R1} = \frac{(V_1 - V_2) + E_1}{R_1}$$

Przykład:



Tu musi być' uziemienie, żeby
admittancja nie była
nieskończona na żółtej
gałęzi. Dzięki takiemu
zabiegowi wiemy, że
 $V_1 = E_1$